

Final Year Project – I Proposal

ON

Ai-Powered Learning System for PDF Based Study

BY:

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A final year project –I proposal submitted to ICS/IT, The University of Agriculture, Peshawar
in partial fulfillment of the requirements for the degree of

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

SESSION: 2022-2026



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Project Overview

The purpose of this project is to develop a Smart AI-Powered Learning System as a web application that helps students study more effectively using PDF-based learning materials. The system is designed to allow users to upload academic PDF documents and interact with them in an organized and intelligent way. Instead of manually reading long documents or using multiple tools, users can access all learning support features in one place.

This learning system mainly focuses on document-based study. Users can read PDFs directly within the application and use AI assistance to better understand the content. The system helps learners engage with their study material by providing intelligent support that improves comprehension and saves time. This approach encourages active learning instead of passive reading.

The system also provides a clean and well-structured interface so users can easily manage documents and focus on their studies without confusion. Overall, this project aims to build a simple, smart, and easy-to-use AI-powered learning platform that helps students' study efficiently, understand concepts clearly, and maintain consistency in their learning process across different devices and environments.

Problem Statement

As digital learning resources continue to increase, many students rely heavily on PDF documents such as lecture notes, books, and academic materials for their studies. However, learning through PDFs is mostly passive and unstructured. Students often struggle to understand complex topics, find relevant information quickly, and stay consistent while studying long documents. As a result, learning becomes time-consuming and less effective.

Additionally, students do not have an interactive way to engage with their PDF materials. If a concept is unclear, they must search externally, which may lead to irrelevant or confusing information. There is also no built-in support for summarization, revision, self-assessment, or progress tracking when studying from documents. To address these issues, there is a need for a smart learning system that can transform static PDF documents into interactive and intelligent study resources. Such a system can help learners understand content more clearly, reduce manual effort, and improve overall learning efficiency.

Objectives

1. To design and develop a smart and user-friendly AI-powered learning system using the MERN Stack that allows users to securely signup, login, and use the system with ease.
2. To provide a structured platform for uploading, managing, and reading PDF documents, enabling learners to organize their study materials efficiently.
3. To integrate artificial intelligence for document-based learning support, including intelligent question-answering, summaries, and concept explanations.
4. To enable learners to improve understanding and revision through AI-generated flashcards and quizzes based on uploaded PDF content.
5. To develop a clean, responsive, and modern user interface that supports document viewing, learning interactions, progress tracking, and works smoothly across different devices.
6. To ensure system security and reliability by implementing authentication, protected routes, and secure handling of user data and uploaded documents.

Key Features & Scope

1. **User Authentication:** Secure signup and login using JWT.
2. **PDF Management:** Upload, organize, and track PDF documents securely.
3. **Embedded PDF Viewer:** View and read documents directly within the application.
4. **AI-Powered Assistance:** After uploading a PDF, users can interact with the same document through contextual AI chat, auto-summary generation, concept explanation, flashcard creation, quiz generation, and access to the complete flashcards list.
5. **Smart Flashcards:** AI-generated flashcards created from document content with a favorites option for important cards. The system will also maintain a centralized flashcards library where users can view, revise, and manage all generated flashcards.
6. **Quiz Generator:** Automatic MCQ generation with configurable question counts and result analysis.
7. **Progress Dashboard:** Displays study activity including documents, flashcards, quizzes, and recent actions.
8. **Responsive UI:** A modern, device-friendly interface built using Tailwind CSS.

Methodology for Implementation

The project will be developed using an agile-like iterative approach, focusing on connecting the frontend, backend, and AI services efficiently.

1. System Architecture & Setup

The development starts with setting up the MERN environment. We will initialize a Node.js and Express backend to handle API requests and a React.js frontend for the user interface. MongoDB will be configured to store user profiles and study data.

2. PDF Processing Workflow

To handle documents, we will implement a file-handling logic where:

- Uploaded PDFs are processed to extract text content.
- The extracted text is then cleaned and formatted so that the AI can read it easily.
- An embedded viewer will be integrated to sync the reading experience with AI features.

3. AI Integration

We will connect the application to the Google Gemini AI API.

- **Prompt Engineering:** We will design specific "prompts" that tell the AI how to behave.
- **Data Flow:** When a user asks a question, the backend sends the relevant PDF text to Gemini, which then sends back an intelligent response.

4. Feature Implementation & Logic

Rather than just building static pages, we will develop the core logic for each interactive tool:

- **Flashcards & Quizzes:** We will write algorithms to convert AI-generated JSON data into interactive UI components like flippable cards and timed quizzes.
- **State Management:** We will use React hooks to manage the "state" of the app, ensuring that when a user marks a favorite or finishes a quiz, the dashboard updates instantly.

5. Deployment & Testing:

Once the features are ready, we will test the system for "Edge Cases". Finally, the application will be optimized for performance using Tailwind CSS to ensure a smooth, lag-free experience on all devices.

System Logic Flowchart

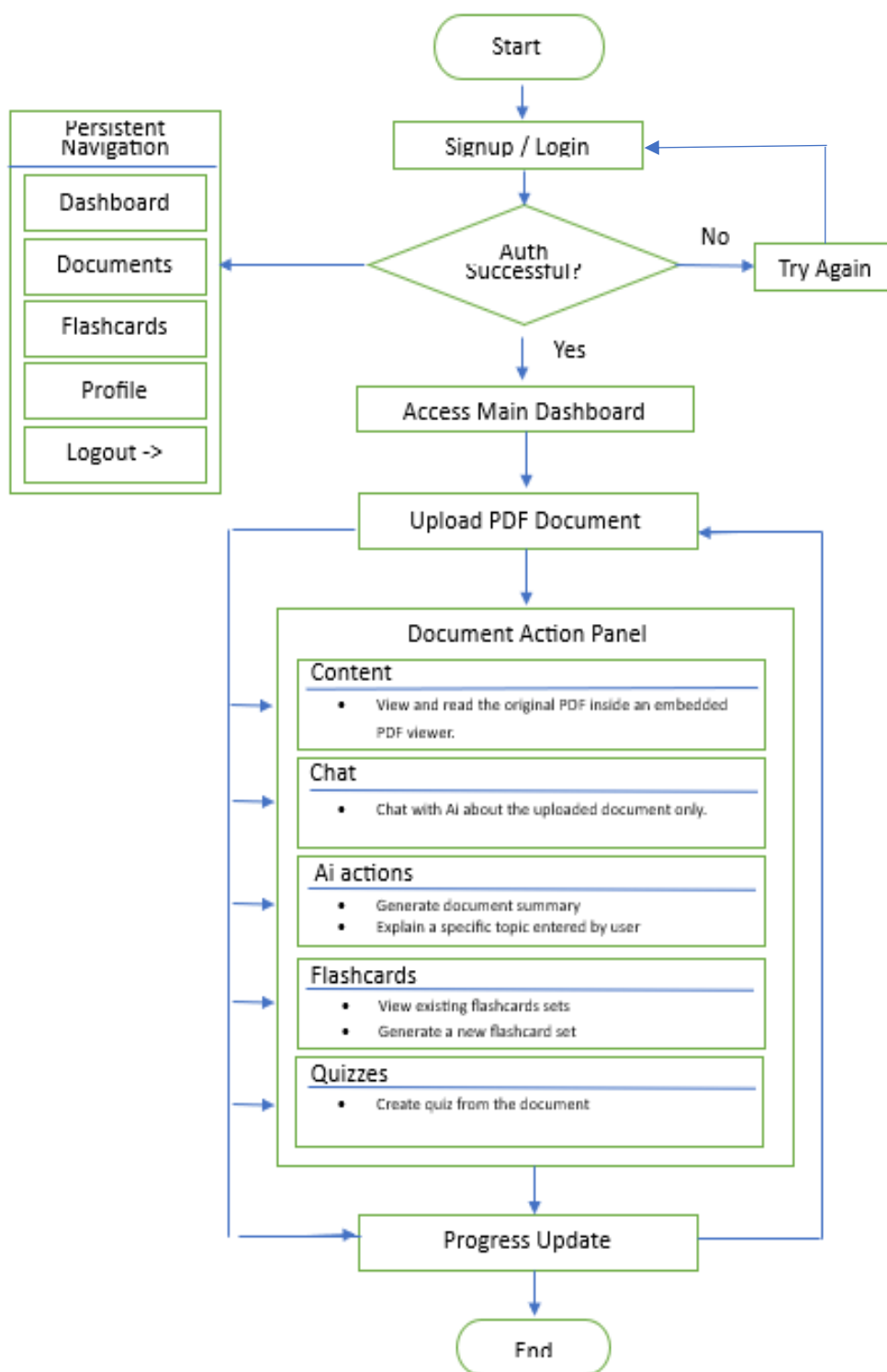


Figure 1 flow chart diagram

Expected Outcome

The successful implementation of the Smart AI-Powered Learning System is expected to deliver a fully functional, intelligent web platform that changes how students interact with PDF materials. The key outcomes include:

1. **Interactive Learning Platform:** A secure web application where students can manage their academic PDFs and study them using AI-powered features in a single interface.
2. **Enhanced Comprehension:** Users will be able to understand complex topics more easily through AI-generated summaries and context-aware concept explanations.
3. **Improved Retention & Revision:** The system will provide automated flashcards and multiple-choice quizzes, helping students test their knowledge and revise efficiently.
4. **Time Efficiency:** Students will save significant time by receiving instant answers to document-related questions instead of searching through external sources.
5. **Progress Visualization:** A comprehensive dashboard will allow users to monitor their study activity, including the number of documents processed and quiz scores achieved.
6. **Responsive User Experience:** A modern, lag-free UI that works consistently across desktops, tablets, and mobile devices, ensuring learning is not restricted to a single environment.

Tools and Technology

The development of the Smart AI-Powered Learning System will utilize a modern and scalable tech stack to ensure high performance and security.

1. Core Development Stack

We will use the MERN stack for a seamless full-stack development experience:

- **MongoDB:** A NoSQL database used to store user profiles, document metadata, and generated study resources.
- **Express.js:** A web application framework for Node.js used to build the backend API.
- **React.js:** A frontend library used to build a dynamic and responsive user interface.
- **Node.js:** A JavaScript runtime environment for executing server-side logic.

2. Artificial Intelligence Integration

- **Google Gemini AI API:** This will serve as the core intelligence engine for PDF analysis, summarization, and quiz generation.
- **Prompt Engineering:** Specialized techniques to optimize AI responses for academic accuracy.

3. Styling and UI/UX

- **Tailwind CSS:** A utility-first CSS framework used to design a clean, modern, and mobile-friendly interface.
- **Lucide React:** For professional and intuitive icons throughout the dashboard.

4. Technical Tools and Libraries

Category	Tool / Library	Purpose
Authentication	JSON Web Token (JWT)	To ensure secure user login and protected routes.
Text Extraction	PDF-Parser	To parse and extract text content from uploaded PDFs for AI processing.
PDF Viewing	React-PDF / PDF.js	For embedding a document viewer directly within the application.
File Uploads	Multer	To handle and store PDF files on the server securely.
API Testing	Postman	To test and debug backend API endpoints during development.
Version Control	Git & GitHub	To manage project source code and track changes.
Deployment (Frontend)	Vercel / Netlify	To host the React frontend for global accessibility.
Deployment (Backend)	Render / Railway	To host the Node.js server and manage live API endpoints.

Expected Timeline (Gantt Chart)

Task	Nov	Dec	Jan	Feb	Mar	Apr
Requirement Analysis & Literature Review	■	■				
System Design (UI & Database Schema)		■	■			
Backend Development (Node, Express & PDF-Parser)			■	■		
AI Integration (Google Gemini API & Prompts)			■	■		
Frontend Development (React & Tailwind)				■	■	
Feature Logic (Flashcards & Quiz Generation)					■	
Testing & Bug Fixing					■	■
Final Documentation & Deployment						■

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